

3- ϕ Ac circuits:-

Introduction:- Single phase system is employed for the operation of almost all the domestic & commercial appliances, e.g. $\rightarrow$ lamps, fans, electric irons, refrigerators, TV sets, washing machine etc.

$\rightarrow$ However, it has its own limitations in the field of Gen^r, transmission, distribution & Industrial applications.

$\rightarrow$ Due to this, it has been Replaced by Polyphase system.

$\rightarrow$ Poly phase (three-phase) system is universally adopted for Gen^r, transmission & distribution of electric power.

Poly phase System:-

Poly $\rightarrow$ means many (more than one) &
phase $\rightarrow$ means windings or circuits, each of them having a single alternating voltage of the same magnitude and frequency.

$\rightarrow$ Hence, a polyphase system is a combination of two or more voltage having same magnitude & frequency, but displaced from one another by equal electrical angle.

$\rightarrow$ This angular displacement between the adjacent voltage is called phase difference & depends upon the number of phases.

(2)

$$\text{phase difference} = \frac{360 \text{ electrical degrees}}{\text{Number of phases}}$$


→ However, the above mentioned relation does not hold good for two-phase system, where the voltages are displaced by 90° electrical.

→ Thus, an AC system having a group of (two or more than two) equal voltages of same frequency arranged to have equal phase difference b/w them is called a poly phase system.

→ The poly phase system may be two-phase system, 3- ϕ system or six-phase system.

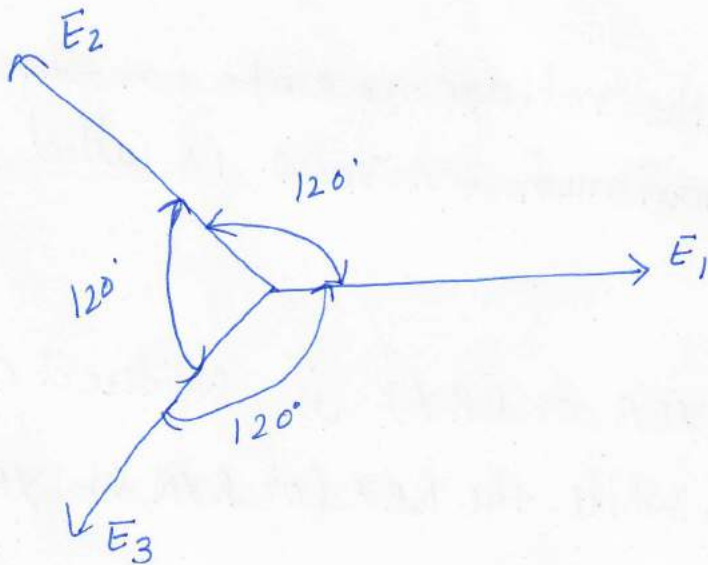
Advantages of 3- ϕ system over single- ϕ system

- 1) Constant Power
- 2) High rating
- 3) power transmission economy
- 4) Superiority of 3- ϕ I.M

Phasor Diagram:-

(3)

→ The emf induced in three coils are of the same magnitude and frequency, but are displaced by 120° (electrical) from each other.



$$E_1 = E_m \sin \omega t, \quad E_2 = E_m \sin(\omega t - 2\pi/3)$$

$$E_3 = E_m \sin(\omega t - 240^\circ)$$

Naming the phases:-

→ The three phases may be Represented by numbers (1, 2 & 3) or by letters (a, b, & c) or by colour (red, yellow & blue, i.e. R, Y, & B).

→ In India, they are named by R, Y & B. that is Red, Yellow & Blue.

Phase Sequence:-

→ In a 3- ϕ system, there are three voltages having same magnitude & frequency displaced by an angle of 120° electrical. They are attaining their ~~their~~ +ve max. value in a particular order.

→ The order in which the voltage or emfs in the three phases attain their maximum +ve value is called the phase sequence.

→ The sequence RYB (or YBR or BRY) is considered as +ve phase sequence, while the RBY (or BYR or YRB) is considered as -ve phase sequence.

The sequence knowledge of phase sequence is essential in the following Important applications:-

- 1) The direction of rotation of 3- ϕ I.M depends upon the phase sequence of 3- ϕ supply. To Reverse the direction of rotation, the phase sequence of the supply given to the Motor has to be changed.
- 2) The parallel operation of three ϕ alternators & transformers is only possible if phase sequence is known.

Double - subscript Notation:-

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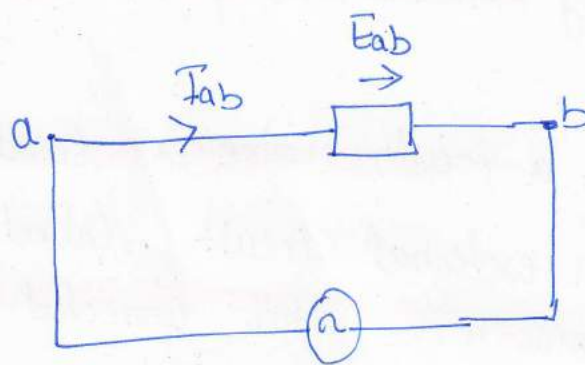


fig → circuit to represent double subscript notation

→ An alternating quantity is generally represented by a double - subscript notation.

→ Current is represented by I_{ab} . It means that

1) The portion ab of the circuit is considered

2) The current flows from a to b .

→ The double - subscript notation is very useful in solving AC circuits having a number of voltages & currents.

Interconnection of three phases:-

1) star or wye (γ) connection

2) Mesh or delta (Δ) connection

1) star or γ connection:-

→ In star or γ connections, the similar ends (either start or finish) of the three windings are connected to a common point called star or neutral point.

→ The three line conductors are run from the remaining three free terminals called line conductors.

→ ordinarily, only three wires are carried to the external circuit giving three-phase, three-wire star connected system.

→ However, sometimes a fourth wire is carried from the star point to the external circuit, called neutral wire, giving three-phase, four wire star-connected system.

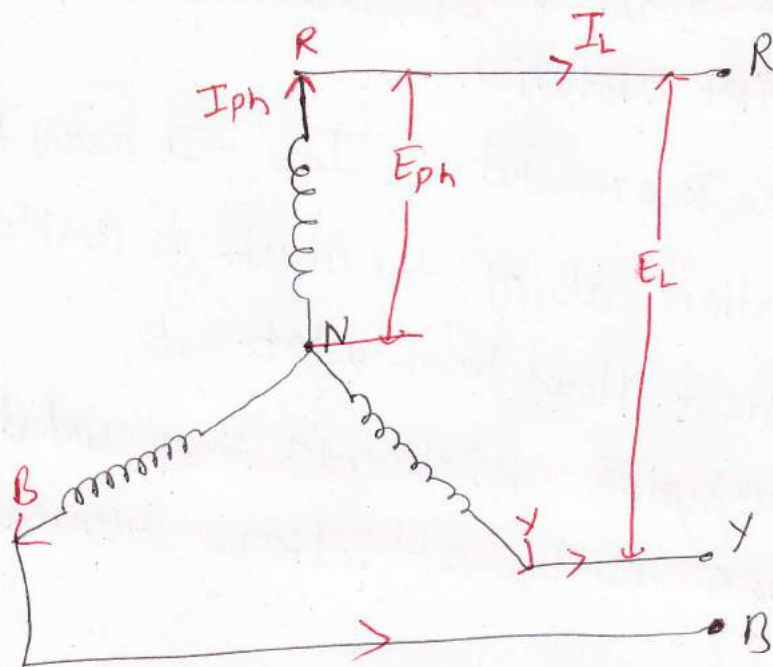


fig ① star connected system

→ three windings are connected to form a star or neutral point.

→ from the remaining three free terminals, ~~the~~ three conductors are run, named R, Y & B.

→ The current flowing through each phase is called Phase Current I_{ph} & current flowing through each line conductor is called line current I_L .

→ Similarly, voltage across each ^{phase} is called phase voltage E_{ph} and voltage across two line conductor is called line voltage (E_L).

1-a) Relation b/w phase voltage & line voltage:-

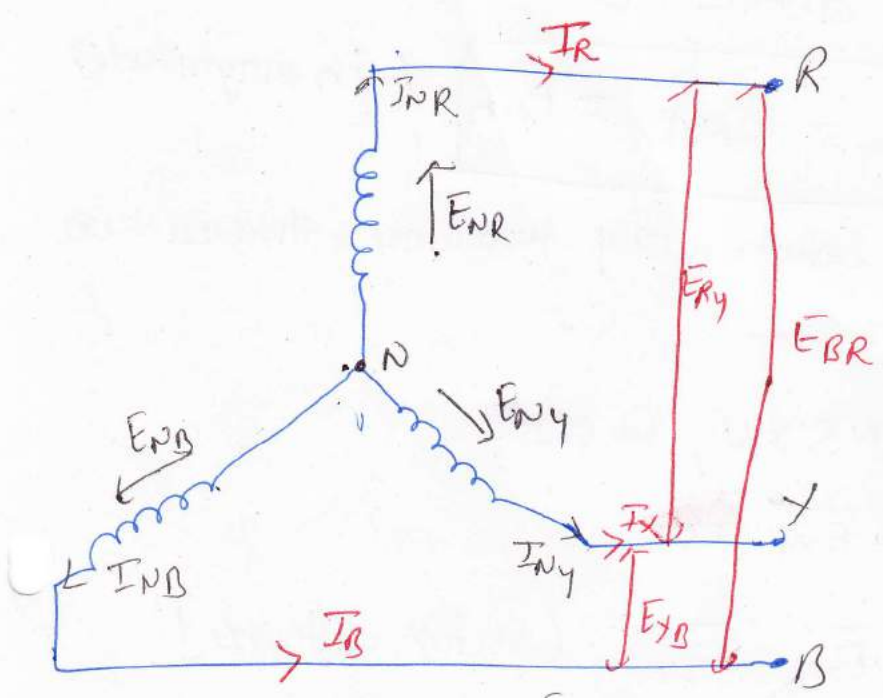


fig → ① circuit
Representing phase
and line voltages &
currents in star
connections.

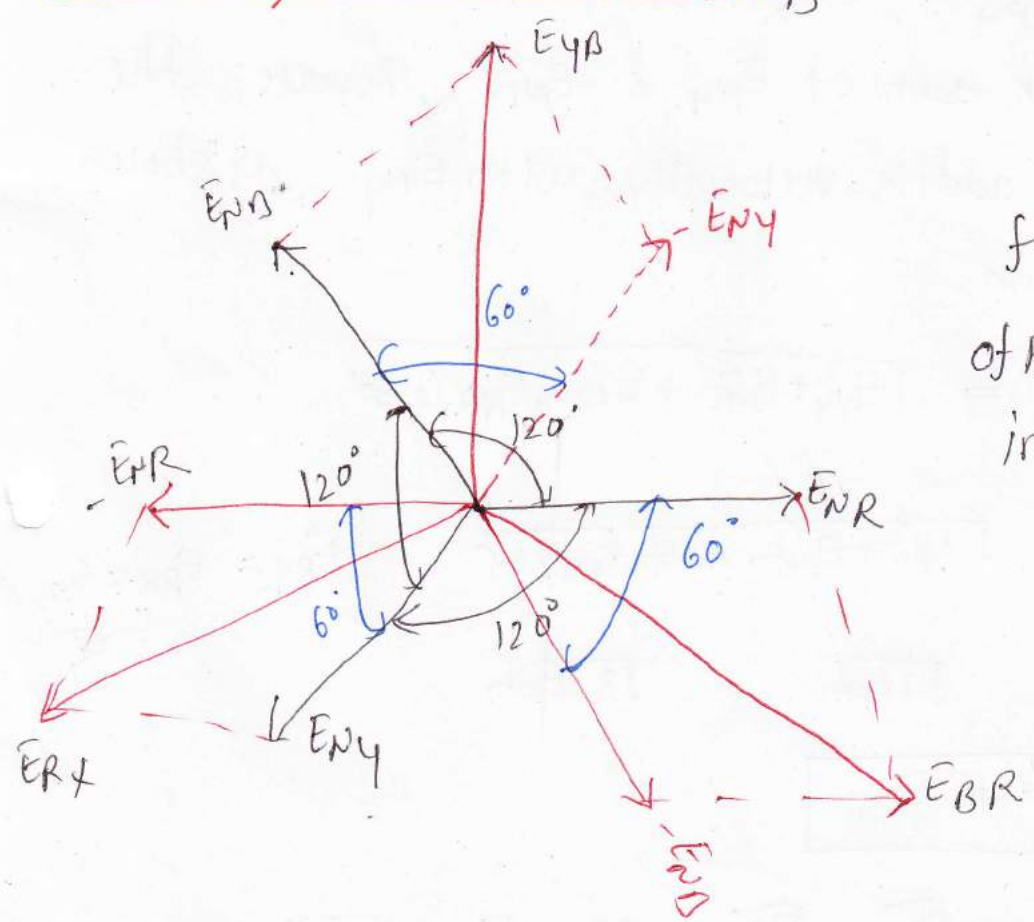


fig → ② phasor diagram
of phase & line voltages
in star connection.

→ The connections are shown in fig 2(a). Since the system is balanced, three voltages E_{NR} , E_{NY} & E_{NB} are equal in magnitude, but displaced from one another by 120° electrical.

→ Their phasor diagrams are shown in fig (B). ⑧

→ The arrow heads of currents indicate +ve direction & not their actual direction at any instant.

Now $\boxed{E_{NR} = E_{NY} = E_{NB} = E_{ph}} \quad (\text{in magnitude})$

→ It may be seen that between any two lines, there are two phase voltages.

Tracing the loop $NRYN$, we get

$$\overline{E}_{NR} + \overline{E}_{RY} - \overline{E}_{NY} = 0$$

$$\overline{E}_{RY} = \overline{E}_{NY} - \overline{E}_{NR} \quad (\text{vector difference})$$

To find the vector sum of $\overline{E}_{NY}$ & $-\overline{E}_{NR}$, Reverse the vector $\overline{E}_{NR}$ and add it vectorially with $\overline{E}_{NY}$, as shown in phasor diagram

$$E_{RY} = \sqrt{E_{NY}^2 + E_{NR}^2 + 2E_{NY}E_{NR}\cos 60^\circ}$$

$$E_L = \sqrt{E_{ph}^2 + E_{ph}^2 + 2E_{ph}E_{ph}\cos 60^\circ}$$

$$\left\{ \begin{aligned} E_{RY} &= E_{BR} = E_{YB} \\ &= E_L \end{aligned} \right.$$

$$E_L = \sqrt{3E_{ph}^2} = \sqrt{3} E_{ph}$$

$$\boxed{E_L = \sqrt{3} E_{ph}}$$

Similarly $\overline{E}_{YB} = \overline{E}_{NB} - \overline{E}_{NY}$ or $E_L = \sqrt{3} E_{ph}$

$$\overline{E}_{BR} = \overline{E}_{NR} - \overline{E}_{NB} = \sqrt{3} E_{ph}$$

Hence, in star connection,

$$\boxed{\text{line voltage} = \sqrt{3} \times \text{phase voltage}}$$

1(b) Relation b/w phase Current & Line Current:-

(9)

from the figure ^{1(b)}, it is clear that same current flows through phase winding as well as the line conductor since line conductor is just connected in series with the phase winding.

$$I_R = I_{NR} \quad , \quad I_Y = I_{NY} \quad \& \quad I_B = I_{NB}$$

Where

$$I_{NR} = I_{NY} = I_{NB} = I_{Ph} \quad (\text{phase current})$$

$$I_R = I_Y = I_B = I_L \quad (\text{line current})$$

In star connection,

$$\begin{array}{c} I_{Ph} = I_L \\ \downarrow \qquad \downarrow \\ (\text{Phase current}) = (\text{line current}) \end{array}$$

② Mesh or Delta Connection:-

→ In delta (Δ) or mesh connections, the finish terminal of one winding is connected to star terminal of the other winding & so on, which forms a closed circuit.

→ The three line conductors are run from three junctions of the mesh called line conductors as shown in fig ②

Relation b/w Phase Current & Line Current:-

(10)

from the

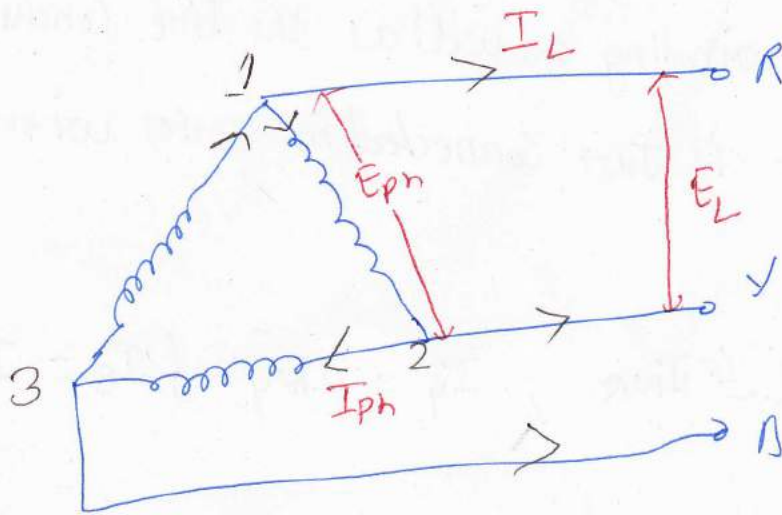


fig (2) Delta connected system.

→ Three conductors R, Y & B are run from the three Junctions called line conductors.

→ The current flowing through each phase is called phase current (I_{ph}) & the current flowing through each line conductor is called line current (I_L)

→ Similarly, voltage across each phase is called phase voltage (E_{ph}) & voltage across two line conductor is called line voltage (E_L).

2-(a) Relation b/w phase voltage & Line Voltage.

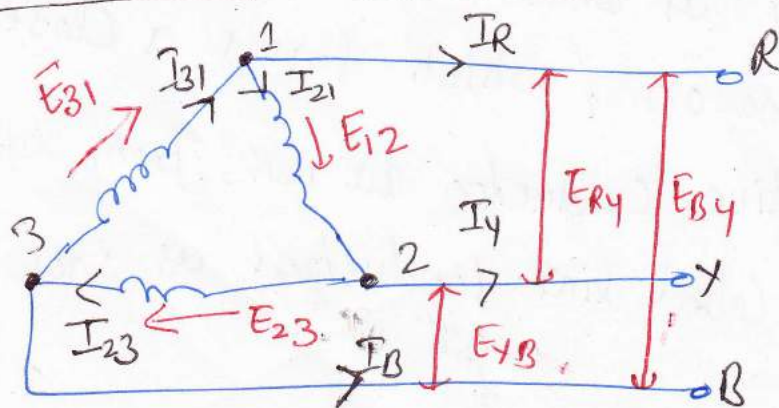


fig → 2(a)
circuit representing
phas & line voltage
& currents in
delta connection

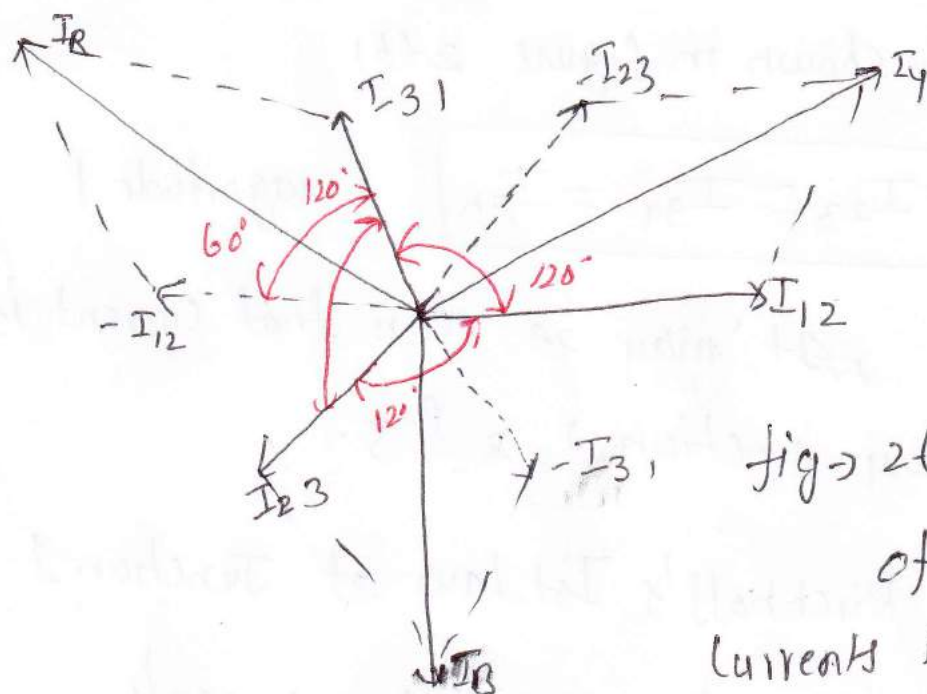


fig 2(s) phasor diagram of phase & line currents in delta connection.

from figure 2-(a), it is clear that voltage across terminals 1 & 2 is the same as across terminals R & Y.

Therefore, $E_{12} = E_{RY}$, similarly, $E_{23} = E_{YB}$ & $E_{31} = E_{BR}$

Where $E_{12} = E_{23} = E_{31} = E_{ph}$ (phase voltage) &

$E_{RY} = E_{YB} = E_{BR} = E_L$ (line voltage)

In delta Connection,

line voltage = phase voltage

$$E_L = E_{ph}$$

2-b) Relation b/w phase current & line current:-

→ Since the system is Balanced, & therefore, three phase currents I_{12} , I_{23} & I_{31} , are equal in magnitude,

but displaced from one another by 120° electrical. (12)

Their phasors are shown in figure 2(b)

$$\boxed{I_{12} = I_{23} = I_{31} = I_{ph}} \quad (\text{magnitude})$$

In fig - 2(a), it may be seen that current is divided at every junction 1, 2 & 3.

→ By Applying Kirchhoff's 1st law at Junction 1.

Incoming currents = outgoing currents

$$\overline{I_{31}} = \overline{I_R} + \overline{I_{12}}$$

$$\Rightarrow \overline{I_R} = \overline{I_{31}} - \overline{I_{12}} \quad (\text{Vector difference})$$

To find the vector sum of I_{31} & $-I_{12}$, reverse the vector I_{12} & add it vectorially with I_{31} as shown in fig 2(b)

$$I_R = \sqrt{I_{12}^2 + I_{31}^2 + 2I_{31}I_{12}\cos 60^\circ}$$

$$I_R = \sqrt{I_{ph}^2 + I_{ph}^2 + 2I_{ph}I_{ph} \times 0.5}$$

$$I_L = \sqrt{3} I_{ph}$$

$$\boxed{I_L = \sqrt{3} I_{ph}}, \quad \text{Similarly, } \overline{I_Y} = \overline{I_{12}} - \overline{I_{23}}$$

Similarly $\overline{I_B} = \overline{I_{23}} - \overline{I_{31}}$ or $\boxed{I_L = \sqrt{3} I_{ph}}$

⊗ Connections of three-phase loads:-

(13)

→ Similar to three-phase supply, the three-phase loads may also be connected in star or delta.

→ The three-phase loads connected in star & delta are shown in fig. below.

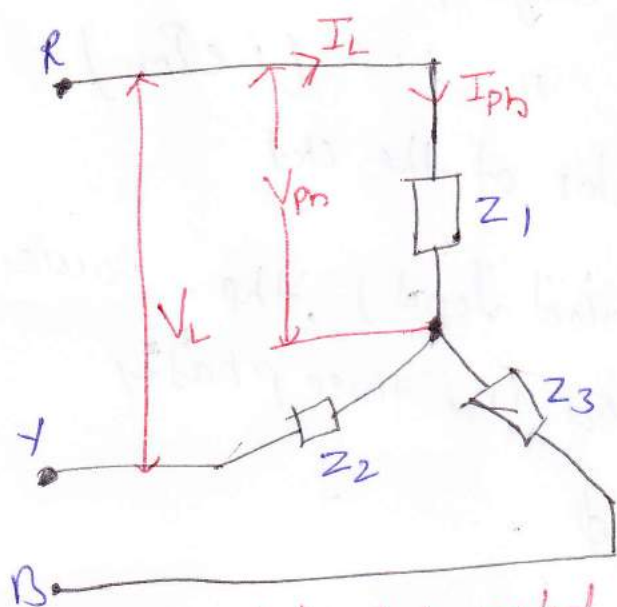


fig (a) → 3- ϕ load connected in star

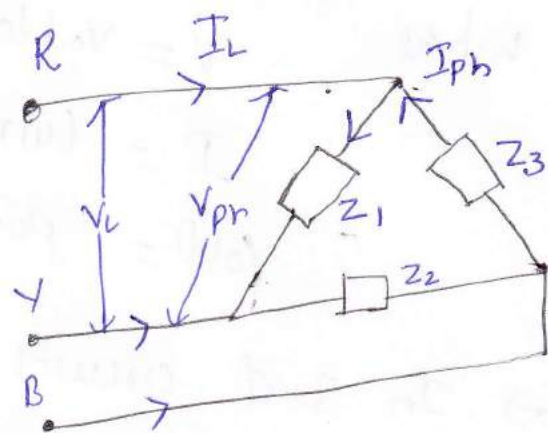


fig (b) 3- ϕ load connected in delta.

→ The 3- ϕ loads may be balanced or unbalanced.

→ If three loads (Impedances) Z_1 , Z_2 & Z_3 are having same magnitude & phase angle, then the 3- ϕ load is said to be a balanced load.

→ Under such condition, all the phase or line currents & all the phase or line voltages are equal in magnitude.

(*) Power in 3- ϕ circuits:-

(14)

→ power in 1- ϕ ~~ckt~~ system or ~~ckt~~ is given by the Relation :-

$$P = V I \cos \phi$$

where

V = voltage of single phase (i.e. V_{ph})

I = current " " " (i.e. I_{ph})

$\cos \phi$ = power factor of the ~~ckt~~

→ In 3- ϕ circuits (balanced load), the power is just the sum of powers in three phases

$$P = 3 V_{ph} I_{ph} \cos \phi$$

~~In star conn~~

~~$P = 3 V_L$~~

In star Connection:-

$$P = 3 \frac{V_L}{\sqrt{3}} I_L \cos \phi$$

$$\left(\because V_{ph} = \frac{V_L}{\sqrt{3}} \right. \\ \left. I_{ph} = I_L \right)$$

$$\boxed{P = \sqrt{3} V_L I_L \cos \phi}$$

In Delta Connection:-

$$P = 3 V_L \frac{I_L}{\sqrt{3}} \cos \phi$$

$$\left(\because V_{ph} = V_L \right. \\ \left. I_{ph} = \frac{I_L}{\sqrt{3}} \right)$$

$$\boxed{P = \sqrt{3} V_L I_L \cos \phi}$$

Apparent power

$$P_a = \sqrt{3} V_L I_L$$

Reactive power

$$P_r = \sqrt{3} V_L I_L \sin \phi$$

⊗ Comparison between star & Delta systems:-

1) Connections:-

Star

Similar ends of the three phases are joined together to get neutral or star point. Line conductor run from the remaining three terminals

Delta

Dissimilar ends of the ~~phas~~ three phases are joined together to form a closed ~~delta~~ delta. Three line conductors run from each joint.

2) Relation b/w Line & phase voltages & currents:-

Star

$$V_L = \sqrt{3} V_{ph}$$

$$I_L = I_{ph}$$

Delta

$$V_L = V_{ph}$$

$$I_L = \sqrt{3} I_{ph}$$

③ Neutral terminal:-

Star

A Neutral wire can be run from the star point

Delta

Neutral wire can-not be Run

④ Voltage to be obtained:-

Star

→ two voltage can be obtained in star-connected four-wire system

- 1) phase voltage b/w any one line conductor & Neutral
- 2) line voltage b/w any two line conductor

Delta

- 1) only one voltage b/w the lines can be obtained.

⑤ Lighting & Power load:-

Star

→ Lighting load can be connected b/w any phase (line conductor) & Neutral while power loads is connected across lines

Delta

→ only power loads can be connected b/w the lines.

Application of Star & Delta Systems:-

(17)

Star

→ These connections are used for domestic (single phase) as well as power (three- ϕ) loads. Protective device (relay, etc.) are connected b/w line & neutral where neutral is properly earthed.

Delta

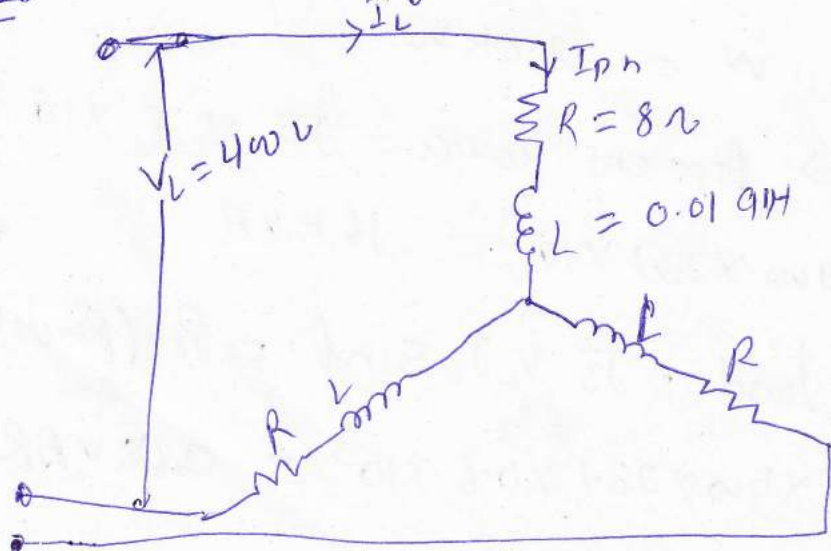
→ These connections are used in transformers for power transmission. Induction Motors are usually connected in delta.

Numericals:-

- (1) Three similar coils each having a Resistance of 8Ω & an Inductance of $0.0191H$ in series in each phase, is connected across a $400V$, three- ϕ $50Hz$ supply. Calculate the line current, power W , KVA & $KVAR$ taken by the load.

Solⁿ

According to Question



$$R = 8\Omega, \quad L = 0.0191\text{ H}, \quad f = 50\text{ Hz}$$

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.0191 = 6\Omega$$

$$V_L = 400\text{ V}$$

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 231\text{ V}$$

$$Z_{ph} = \sqrt{R^2 + X_L^2} = \sqrt{8^2 + 6^2}$$

$$Z_{ph} = 10\Omega$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{231}{10} = 23.1\text{ A}$$

$$I_L = I_{ph} = 23.1\text{ A}$$

$$\cos \phi = \frac{R_{ph}}{Z_{ph}} = \frac{8}{10} = 0.8 \text{ lagging}$$

$$P_{in} = \sqrt{3} V_L I_L \cos \phi = \sqrt{3} \times 400 \times 23.1 \times 0.8$$

$$P_{in} = 12,800\text{ W} = 12.8\text{ kW}$$

KVA taken by the load $\rightarrow$ Apparent power $= \sqrt{3} V_L I_L \times 10^{-3}$

$$P_a = \sqrt{3} \times 400 \times 23.1 \times 10^{-3} = 16\text{ KVA}$$

KVAR taken by the load $= \sqrt{3} V_L I_L \sin \phi = P_r$ (Reactive Power)

$$P_r = \sqrt{3} \times 400 \times 23.1 \times 0.6 \times 10^{-3} = 9.6\text{ KVAR}$$

② A Balanced 3- ϕ star-connected load of 18 kW taking a leading current of 60 A, when connected across a 3- ϕ , 440 V, 50 Hz supply. Find the values & the nature of the load.

Solⁿ According to Question

Given that

load supplied, $P = 18 \text{ kW} = 18,000 \text{ W}$

Line voltage, $V_L = 440 \text{ V}$

Line current, $I_L = 60 \text{ A}$

$$\text{Phase voltage, } V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{440}{\sqrt{3}} = \frac{230.94 \text{ V}}{\sqrt{3}} = 254 \text{ V}$$

$$I_{ph} = I_L = 60 \text{ A}$$

$$Z_{ph} = \frac{V_{ph}}{I_{ph}} = \frac{254}{60} = 4.234 \Omega$$

$$\cos \phi = \frac{P}{\sqrt{3} V_L I_L} = \frac{18000}{\sqrt{3} \times 440 \times 60} = 0.39365$$

(leading)

($\because$ current is leading)

$$R = Z \cos \phi \quad (\text{Resistance of the load})$$

$$R = 4.234 \times 0.39365 = 1.6667 \Omega$$

$$\text{Reactance of load, } X = Z \sin \phi$$

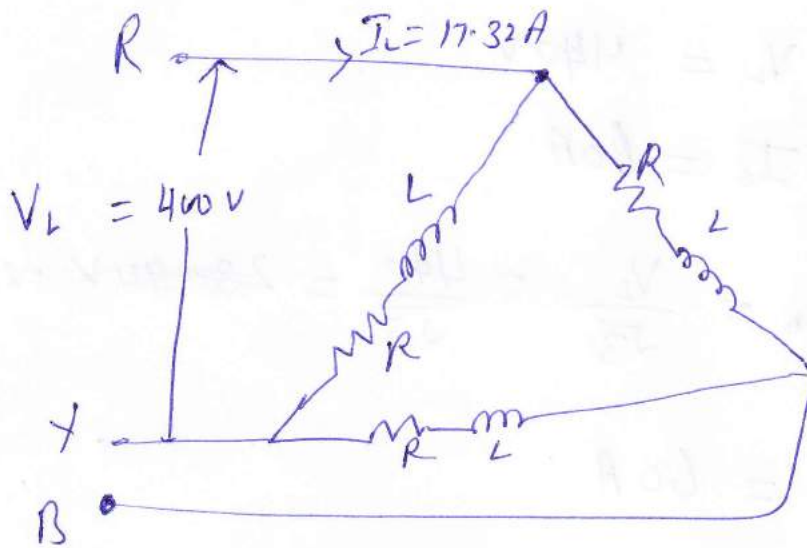
$$X = 4.234 \times 0.9193 = 3.89242 \quad (\text{capacitive})$$

($\because$ current is leading)

② Three identical coils connected in delta across 400 V, 50 Hz, 3- ϕ AC supply, taken a line current of 17.32 A at power factor of 0.8 lagging. Calculate i) the phase current (ii) the Resistance & Inductance of each coil iii) the Power drawn by the each coil.

Solⁿ

$$V_{ph} = V_L = 400V$$



$$I_L = 17.32 A$$

$$\cos \phi = 0.8 \text{ lag}$$

$$1) \quad I_{ph} = \frac{I_L}{\sqrt{3}} = \frac{17.32}{\sqrt{3}} = 10 A$$

$$Z_{ph} = \frac{V_{ph}}{I_{ph}} = \frac{400}{10} = 40 \Omega$$

$$2) \quad R_{ph} = Z_{ph} \cos \phi = 40 \times 0.8 = 32 \Omega$$

$$X_{ph} = \sqrt{Z_{ph}^2 - R_{ph}^2} = \sqrt{40^2 - 32^2} = 24 \Omega$$

$$L = \frac{X_{ph}}{2\pi f} = \frac{24}{2\pi \times 50} = 0.0764 H = 76.4 mH$$

$$3) \quad P = I_{ph}^2 R_{ph} = 10^2 \times 32 = 3200 W$$